

HDOC - Day 11

Digit Operations Using while, for and do-while Loops

Problem

Write a menu-driven C program using switch-case to input a positive integer and perform: (1) sum of squares of digits using a while loop, (2) sum of cubes using a for loop, or (3) difference between the sum of cubes and sum of squares using a do-while loop.

Algorithm

1. Start. 2. Read a positive integer n; if n is not positive, request it again. 3. Display the menu and read choice. 4. Choice 1: copy n to temp, set sum=0, and use a while loop to extract each digit with temp%10, add digit*digit, and remove the digit using temp/10. 5. Choice 2: use a for loop with temp=n; temp>0; temp/=10 and add digit*digit*digit. 6. Choice 3: set squareSum=0 and cubeSum=0, then use a do-while loop to calculate both sums; display cubeSum-squareSum. 7. Display an error for invalid choice. 8. Repeat menu until Exit. 9. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n=123; choice=1	Sum of squares = 14	Sum of squares = 14	Pass
2	n=123; choice=2	Sum of cubes = 36	Sum of cubes = 36	Pass
3	n=123; choice=3	Difference = 22	Difference = 22	Pass
4	n=405; choices 1,2	Squares=41; Cubes=189	Squares=41; Cubes=189	Pass
5	choice=9	Invalid choice	Invalid choice	Pass

C Program

```
#include <stdio.h>
int main() {
    int n, choice, temp, digit;
    int sum, squareSum, cubeSum;

    do {
        printf("Enter a positive integer: ");
        scanf("%d", &n);
        if(n <= 0) printf("Invalid input. Enter a positive integer.\n");
    } while(n <= 0);

    do {
        printf("\n1. Sum of squares of digits\n");
        printf("2. Sum of cubes of digits\n");
        printf("3. Difference (cubes - squares)\n");
        printf("4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                temp = n; sum = 0;
                while(temp > 0) {
                    digit = temp % 10;
                    sum += digit * digit;
                    temp /= 10;
                }
                printf("Sum of squares = %d\n", sum);
                break;

            case 2:
                sum = 0;
                for(temp = n; temp > 0; temp /= 10) {
                    digit = temp % 10;
                    sum += digit * digit * digit;
                }
                printf("Sum of cubes = %d\n", sum);
                break;

            case 3:
                temp = n; squareSum = 0; cubeSum = 0;
                do {
                    digit = temp % 10;
                    squareSum += digit * digit;
                    cubeSum += digit * digit * digit;
                    temp /= 10;
                } while(temp > 0);
                printf("Difference = %d\n", cubeSum - squareSum);
                break;

            case 4:
                break;

            default:
                printf("Invalid choice.\n");
        }
    } while(choice != 4);

    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang digits.c -o digits
puppie@Anvikas-MacBook-Air Desktop % ./digits
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cubes - squares)
4. Exit
Enter choice: 1
Sum of squares = 14
```

```
puppie@Anvikas-MacBook-Air Desktop % ./digits
Enter a positive integer: 123
Enter choice: 2
Sum of cubes = 36
```

```
puppie@Anvikas-MacBook-Air Desktop % ./digits
Enter a positive integer: 123
Enter choice: 3
Difference = 22
```

```
puppie@Anvikas-MacBook-Air Desktop % ./digits
Enter a positive integer: 405
Enter choice: 1
Sum of squares = 41
Enter choice: 2
Sum of cubes = 189
```

Result: The menu-driven digit-operation program was compiled, executed and tested successfully. The while loop, for loop and do-while loop produced the expected results for the prepared test cases.